

A Detailed Formalisation Blueprint

From current Mathlib to Grothendieck's Coherence Theorem
(EGA III, Théorème 3.2.1)

Blueprint for a Lean 4 / Mathlib formalisation

July 6, 2026

Contents

I	Baseline, strategy, and the target	3
1	The target theorem	4
2	What Mathlib provides today	5
3	What must be built, and the critical path	7
4	Conventions	8
II	Quasi-coherent sheaves on schemes	9
5	The affine model: the functor $\widetilde{(-)}$	10
6	Quasi-coherent sheaves on general schemes	11
7	Invertible sheaves and twists	13
III	Cohomology of $\mathcal{O}_X$ -modules	14
8	Sheaf cohomology and higher direct images	16
9	Flasque sheaves and acyclic resolutions	17
10	Comparison: abstract rightDerived vs sheaf cohomology	19
IV	Čech cohomology and vanishing on affines	20
11	The Čech complex	22
12	Vanishing on affines and the comparison	23
13	Higher direct images of quasi-coherent sheaves	24
V	Projective space and Proj	25
14	Twisting sheaves and graded modules on Proj	27

15 Cohomology of projective space	28
16 Serre finiteness and vanishing on Proj	29
VI Coherent sheaves	30
17 Coherence on locally Noetherian schemes	32
18 Closed immersions and support	33
VII Projective morphisms	34
19 Ample invertible sheaves and projective morphisms	36
20 Higher direct images along projective morphisms	37
VIII Dévissage	38
21 Dévissage of coherent sheaves	40
IX Chow's lemma	41
22 Chow's lemma	43
X The finiteness theorem	44
23 Assembly	45
XI The target Lean statement	46
24 From coherence to finite presentation	47
XII Dependency graph, critical path, and work packages	48
25 Critical path and packaging	49
25.1 Critical path to Theorem 23.1	49
25.2 Suggested work packages (parallelisable)	49
25.3 Largest single items	50
25.4 Items closest to existing Mathlib	50

Part I

Baseline, strategy, and the target

Chapter 1

The target theorem

Theorem 1.1 (Grothendieck’s Coherence Theorem; EGA III, Théorème 3.2.1). *Let Y be a locally Noetherian scheme and $f: X \rightarrow Y$ a proper morphism. For every coherent $\mathcal{O}_X$ -module $\mathcal{F}$ and every $n \geq 0$, the $\mathcal{O}_Y$ -module $R^n f_* \mathcal{F}$ is coherent. Equivalently, in the form of `derivedpushforward.lean`: for f proper, Y locally Noetherian and M a sheaf of modules of finite presentation on X , the object $((\text{Modules.pushforward } f).\text{rightDerived } n)(M)$ is of finite presentation for every n .*

Lean: `AlgebraicGeometry.Scheme.rightDerivedPushforward_isFinitePresentation_of_isProper`

Uses: `thm:finiteness`, `prop:bridge-rderived`, `lem:fp-iff-coh`

[absent from Mathlib: to be built]

Reference: EGA III [2]; Stacks, Proposition 30.19.1, tag 0205 [8]; Hartshorne, III.8.8 (projective case) [3]; Vakil, §18.9 [4]. The whole blueprint is the proof of Theorem 1.1: Part II–IV build the missing cohomological substrate on top of what Mathlib has today, Parts V–IX build the geometric inputs, and Part X assembles the theorem.

Chapter 2

What Mathlib provides today

The following are present in Mathlib and are the starting line. Item names are the relevant modules or declarations; “modulo API glue” means the object exists but auxiliary lemmas about it may still be needed.

Schemes. `AlgebraicGeometry.Scheme`, morphisms, `Spec`, the $\Gamma \dashv \text{Spec}$ adjunction, fibre products, open/closed immersions, and a large morphism-property library (finite type, separated, quasi-compact, quasi-separated, affine, universally closed, integral, closed immersion).

[in Mathlib today, modulo API glue]

Proper and Noetherian. `AlgebraicGeometry.Morphisms.Proper (IsProper)`; `AlgebraicGeometry.Noetherian (IsNoetherian, IsLocallyNoetherian)`.

[in Mathlib today, modulo API glue]

Sheaves of modules. `Mathlib.Algebra.Category.ModuleCat.Sheaf` and `AlgebraicGeometry.Modules.Sheaf`: the category `SheafOfModules`, its abelian structure (`Modules.instAbelian`), the pushforward and pullback functors (`Scheme.Modules.pushforward`, `pullback`), the adjunction (`pullbackPushforwardAdjunction`) and the coherence isomorphisms `pushforwardId`, `pushforwardComp`.

[in Mathlib today, modulo API glue]

Quasi-coherence, abstractly. `Mathlib.Algebra.Category.ModuleCat.Sheaf.Quasicohherent`: a sheaf of modules is quasi-coherent if it is locally the cokernel of a map between coproducts of copies of the structure sheaf, and of finite presentation when these are finite (`SheafOfModules.IsFinitePresentation`).

[in Mathlib today, modulo API glue]

The affine model $\widetilde{M}$. `AlgebraicGeometry.Modules.Tilde`: for an R -module M , the sheaf $\widetilde{M}$ of $\mathcal{O}_{\text{Spec } R}$ -modules, with its localisation-to-stalk maps.

[partly in Mathlib: must be extended] (the sheaf and its stalks exist; the equivalence $\text{QCoh}(\text{Spec } R) \simeq \text{Mod}_R$ and the pushforward/pullback compatibilities are not all present).

Derived functors. `CategoryTheory.Abelian.RightDerived` and `Abelian.GrothendieckCategory.EnoughInjectives`: any Grothendieck abelian category (in particular `SheafOfModules`) has enough injectives, so `Functor.rightDerived` of a left exact functor is defined; hence `(Modules.pushforward f).rightDerived n` exists.

[in Mathlib today, modulo API glue]

Proj. `AlgebraicGeometry.ProjectiveSpectrum.Scheme`: $\text{Proj } S$ of a graded ring as a scheme.

[partly in Mathlib: must be extended] (the scheme exists; the twisting sheaves and the graded-module $\rightarrow$ sheaf functor are absent).

Homological algebra. Abelian categories, injective resolutions, δ -functors, long exact sequences, Ext, spectral objects and some spectral-sequence API (`CategoryTheory.Abelian`, `Homology`).

[in Mathlib today, modulo API glue]

Chapter 3

What must be built, and the critical path

Relative to Chapter 2, the development requires the following, in dependency order. Each is a Part below.

1. **Quasi-coherent sheaves on schemes** (Part II): the affine equivalence via $\widetilde{(-)}$, quasi-coherence of the geometric $\mathcal{O}_X$ -modules, and the operations used later.
2. **Cohomology of $\mathcal{O}_X$ -modules** (Part III): $H^i(X, -)$ and $R^i f_*$ as derived functors, the flasque package, and the comparison of the abstract `rightDerived` with geometric cohomology.
3. **Čech cohomology and affine vanishing** (Part IV): the Čech complex, Serre’s vanishing on affines, Čech computes cohomology for affine covers, and quasi-coherence of $R^i f_*$.
4. **Projective space and Proj** (Part V): twisting sheaves, coherent sheaves from graded modules, and the explicit cohomology of $\mathbf{P}^n$.
5. **Coherent sheaves** (Part VI): the coherence predicate, coherence = finite presentation on locally Noetherian schemes, operations, closed immersions, support.
6. **Projective morphisms** (Part VII): relative ampleness and coherence/vanishing of $R^p f_*$ along projective morphisms.
7. **Dévissage** (Part VIII) and **Chow’s lemma** (Part IX).
8. **The finiteness theorem** (Part X) and the **bridge** to the Lean statement (Part XI).

Remark 3.1 (On spectral sequences). The relative Leray (Grothendieck) spectral sequence is *not* on the critical path. Following the note [7] incorporated earlier, the single step of the [EGA] proof that used it is carried out by the flasque acyclic-pushforward Lemma 9.6, which lives in the flasque layer of Part III and needs only: injectives are flasque, pushforward preserves flasqueness, flasque sheaves are acyclic, and acyclic resolutions compute derived functors. No construction of a spectral sequence of a composite of functors is required anywhere below.

Chapter 4

Conventions

Schemes and morphisms are as in Mathlib’s `AlgebraicGeometry`. For an $\mathcal{O}_X$ -module $\mathcal{F}$, $H^i(X, \mathcal{F})$ is the i -th right derived functor of $\Gamma(X, -)$ on the abelian category of $\mathcal{O}_X$ -modules, and $R^i f_*$ the i -th right derived functor of f_* . “Coherent” is Definition 17.1; on locally Noetherian schemes it coincides with quasi-coherent of finite type and with finite presentation (Lemma 17.4). A *projective morphism* is one admitting a closed immersion into some $\mathbf{P}_S^n$ over the base (Definition 19.1).

Part II

Quasi-coherent sheaves on schemes

Chapter 5

The affine model: the functor $\widetilde{(-)}$

Reference: Stacks, Section 26.7, tag [01I6](#), and Section 17.10 [\[8\]](#); Hartshorne, II.5 [\[3\]](#).

Construction 5.1 (The functor $\widetilde{(-)}$). For a commutative ring R , sending an R -module M to the $\mathcal{O}_{\text{Spec } R}$ -module $\widetilde{M}$ (Mathlib’s `ModuleCat.tildeInModuleCat`) is functorial and additive, giving $\widetilde{(-)}: \text{Mod}_R \rightarrow \text{Mod}(\mathcal{O}_{\text{Spec } R})$.

Lean: `ModuleCat.tildeFunctor`

[partly in Mathlib: must be extended]

Lemma 5.2 (Sections and stalks of $\widetilde{M}$). For $f \in R$ and a prime $\mathfrak{p}$, $\widetilde{M}(D(f)) \cong M_f$ (as R_f -modules), $\Gamma(\text{Spec } R, \widetilde{M}) \cong M$, and $(\widetilde{M})_{\mathfrak{p}} \cong M_{\mathfrak{p}}$; these are natural in M . Reference: Stacks, Lemma 26.7.1, tag [01I6](#).

Lean: `ModuleCat.Tilde.sections_basicOpen`, `ModuleCat.Tilde.stalkIso`

Uses: `con:tilde`

[partly in Mathlib: must be extended]

Lemma 5.3 ($\widetilde{(-)}$ is exact and fully faithful). The functor $\widetilde{(-)}$ is exact and fully faithful; on morphisms it is the bijection $\text{Hom}_R(M, N) \cong \text{Hom}_{\mathcal{O}_{\text{Spec } R}}(\widetilde{M}, \widetilde{N})$. Reference: Stacks, tag [01I6](#); Hartshorne, II.5.2, II.5.5.

Lean: `ModuleCat.tildeFunctor_fullyFaithful`, `ModuleCat.tildeFunctor_exact`

Uses: `lem:tilde-sections`

[absent from Mathlib: to be built]

Proposition 5.4 (Affine equivalence). $\widetilde{(-)}$ induces an equivalence $\text{Mod}_R \simeq \text{QCoh}(\text{Spec } R)$ onto the quasi-coherent $\mathcal{O}_{\text{Spec } R}$ -modules, with quasi-inverse $\Gamma(\text{Spec } R, -)$. Reference: Stacks, Lemma 26.7.1 and Section 26.7, tag [01I6](#).

Lean: `AlgebraicGeometry.Spec.qcohEquivModule`

Uses: `lem:tilde-ff`, `def:qcoh`

[absent from Mathlib: to be built]

Lemma 5.5 (Functoriality on affines). For a ring map $\varphi: R \rightarrow S$ inducing $\psi: \text{Spec } S \rightarrow \text{Spec } R$: $\psi_* \widetilde{N} \cong \widetilde{N}_R$ (restriction of scalars) for an S -module N , and $\psi^* \widetilde{M} \cong \widetilde{S \otimes_R M}$ for an R -module M , naturally. Reference: Stacks, Lemma 26.7.3, tag [01I6](#).

Lean: `AlgebraicGeometry.Spec.pushforward_tilde`, `AlgebraicGeometry.Spec.pullback_tilde`

Uses: `prop:affine-equiv`

[absent from Mathlib: to be built]

Chapter 6

Quasi-coherent sheaves on general schemes

Definition 6.1 (Quasi-coherent $\mathcal{O}_X$ -module). An $\mathcal{O}_X$ -module $\mathcal{F}$ is *quasi-coherent* if X has an affine open cover $\{U_i = \text{Spec } R_i\}$ with $\mathcal{F}|_{U_i} \cong \widetilde{M_i}$ for R_i -modules M_i . Reference: Stacks, tag [01BD](#); Hartshorne, II.5.4.

Lean: `AlgebraicGeometry.Scheme.Modules.IsQuasicoherent`

Uses: `con:tilde`

[absent from Mathlib: to be built]

Lemma 6.2 (Agreement with the abstract predicate). *Definition 6.1 agrees with the abstract quasi-coherence of `SheafOfModules.IsQuasicoherent` (local presentation as a cokernel of free modules): a geometric $\mathcal{O}_X$ -module is quasi-coherent in the sense of Definition 6.1 iff its image in `SheafOfModules` is. This is the link between the geometric layer and the existing Mathlib predicate. Reference: Stacks, tag [01BG](#).*

Lean: `AlgebraicGeometry.Scheme.Modules.isQuasicoherent_iff_sheafOfModules`

Uses: `def:qcoh`, `prop:affine-equiv`

[absent from Mathlib: to be built]

Lemma 6.3 (Local nature). *Quasi-coherence is local: if $\mathcal{F}|_{U_i}$ is quasi-coherent for an open cover $\{U_i\}$ then $\mathcal{F}$ is; and on an affine $\text{Spec } R$ every quasi-coherent sheaf is $\widetilde{M}$ for $M = \Gamma$. Reference: Stacks, tag [01I6](#).*

Lean: `AlgebraicGeometry.Scheme.Modules.isQuasicoherent_of_cover`

Uses: `def:qcoh`, `prop:affine-equiv`

[absent from Mathlib: to be built]

Lemma 6.4 (The structure sheaf and pullbacks). *$\mathcal{O}_X$ is quasi-coherent, and for any $f: X \rightarrow Y$ the pullback $f^*\mathcal{G}$ of a quasi-coherent $\mathcal{G}$ is quasi-coherent. Reference: Stacks, tags [01BE](#), [01BM](#).*

Lean: `AlgebraicGeometry.Scheme.Modules.isQuasicoherent_pullback`

Uses: `def:qcoh`, `lem:tilde-functorial`

[absent from Mathlib: to be built]

Proposition 6.5 (Quasi-coherent sheaves form a Serre–abelian subcategory). *Kernels, cokernels, images and finite direct sums of morphisms of quasi-coherent sheaves are quasi-coherent, as are extensions; hence $\text{QCoh}(X)$ is a weak Serre subcategory of $\text{Mod}(\mathcal{O}_X)$ and is abelian, and the inclusion is exact. Reference: Stacks, tags [01BJ](#), [01LA](#).*

Lean: AlgebraicGeometry.Scheme.Modules.qcoh_isAbelian
Uses: def:qcoh, lem:tilde-ff, prop:affine-equiv
[absent from Mathlib: to be built]

Remark 6.6. Quasi-coherence of *pushforwards* f_* and of $R^i f_*$ needs cohomological input (vanishing on affines) and is deferred to Chapter [13](#).

Chapter 7

Invertible sheaves and twists

Only a light amount is needed, for the projective case and Chow’s lemma.

Definition 7.1 (Locally free and invertible). An $\mathcal{O}_X$ -module is *locally free of rank r* if X has an open cover on which it is $\mathcal{O}^{\oplus r}$, and *invertible* if locally free of rank 1. Invertible sheaves form a group $\text{Pic}(X)$ under $\otimes_{\mathcal{O}_X}$. Reference: Stacks, tag [01CR](#); Hartshorne, II.6.12.

Lean: `AlgebraicGeometry.Scheme.Modules.IsInvertible`

[partly in Mathlib: must be extended]

Lemma 7.2 (Tensoring by an invertible sheaf is exact). *For an invertible $\mathcal{L}$, the functor $- \otimes_{\mathcal{O}_X} \mathcal{L}$ is exact and preserves quasi-coherence and (later) coherence, with inverse $- \otimes \mathcal{L}^{-1}$. Reference: Stacks, tag [01CR](#).*

Lean: `AlgebraicGeometry.Scheme.Modules.tensorInvertible_exact`

Uses: `def:invertible`, `prop:qcoh-abelian`

[absent from Mathlib: to be built]

Part III

Cohomology of $\mathcal{O}_X$ -modules

References: Stacks, Chapter 20 (Cohomology of Sheaves) and Chapter 30 (Cohomology of Schemes) [8]; Hartshorne, III.1–III.2, III.8 [3].

Chapter 8

Sheaf cohomology and higher direct images

Definition 8.1 (Sheaf cohomology). For a scheme X , $H^i(X, -)$ is the i -th right derived functor of $\Gamma(X, -): \text{Mod}(\mathcal{O}_X) \rightarrow \text{Ab}$, computed by injective resolutions (which exist: $\text{Mod}(\mathcal{O}_X)$ is a Grothendieck abelian category, already available in Mathlib). Reference: Stacks, tag [01FR](#); Hartshorne, III.2.

Lean: `AlgebraicGeometry.Scheme.cohomology`

Uses:

[absent from Mathlib: to be built]

Definition 8.2 (Higher direct image). For $f: X \rightarrow Y$, $R^i f_*$ is the i -th right derived functor of the left exact functor $f_*: \text{Mod}(\mathcal{O}_X) \rightarrow \text{Mod}(\mathcal{O}_Y)$. Reference: Stacks, tag [01F1](#); Hartshorne, III.8.

Lean: `AlgebraicGeometry.Scheme.higherDirectImage`

Uses:

[absent from Mathlib: to be built]

Lemma 8.3 (Left exactness). $\Gamma(X, -)$ and f_* are additive and left exact; f_* is right adjoint to f^* (present in Mathlib as `pullbackPushforwardAdjunction`), whence left exactness. Reference: Stacks, tag [01F1](#).

Lean: `AlgebraicGeometry.Scheme.Modules.pushforward_leftExact`

Uses: `def:higher-direct-image`

[partly in Mathlib: must be extended]

Lemma 8.4 (Cohomological δ -functor). $\{H^i(X, -)\}$ and $\{R^i f_*\}$ are cohomological δ -functors: a short exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ yields a natural long exact sequence, with $H^0 = \Gamma$, $R^0 f_* = f_*$. Reference: Stacks, tag [0156](#); the δ -functor API is in Mathlib (`CategoryTheory.Abelian.RightDerived`).

Lean: `AlgebraicGeometry.Scheme.higherDirectImage_deltaFunctor`

Uses: `def:sheaf-cohomology`, `def:higher-direct-image`, `lem:pf-left-exact`

[partly in Mathlib: must be extended]

Lemma 8.5 (Higher direct image as a sheafification). $R^i f_* \mathcal{F}$ is the sheaf associated to the presheaf $V \mapsto H^i(f^{-1}(V), \mathcal{F}|_{f^{-1}(V)})$ on Y . In particular $R^i f_* \mathcal{F}|_V$ depends only on $\mathcal{F}|_{f^{-1}(V)}$, so the computation of $R^i f_*$ is local on Y . Reference: Stacks, Lemma 20.7.3, tag [01E0](#); Hartshorne, III.8.1.

Lean: `AlgebraicGeometry.Scheme.higherDirectImage_isSheafification`

Uses: `def:sheaf-cohomology`, `def:higher-direct-image`

[absent from Mathlib: to be built]

Chapter 9

Flasque sheaves and acyclic resolutions

This chapter is the light replacement for the relative Leray spectral sequence (Remark 3.1); it also supplies the acyclic resolutions used to compute cohomology in later parts.

Definition 9.1 (Flasque sheaf). A sheaf $\mathcal{F}$ on a topological space is *flasque* if $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is surjective for all opens $V \subseteq U$. Reference: Stacks, tag 01AF; Hartshorne, II. Ex.1.16.

Lean: `TopCat.Presheaf.IsFlasque`
[absent from Mathlib: to be built]

Lemma 9.2 (Injectives are flasque). *Every injective $\mathcal{O}_X$ -module is flasque. Reference: Stacks, tag 01EA; Hartshorne, III.2.4.*

Lean: `AlgebraicGeometry.Scheme.Modules.injective_isFlasque`
Uses: `def:flasque`
[absent from Mathlib: to be built]

Lemma 9.3 (Pushforward preserves flasqueness). *If $\mathcal{F}$ is flasque on X and $g: X \rightarrow X'$ continuous, then $g_*\mathcal{F}$ is flasque, since $(g_*\mathcal{F})(V) = \mathcal{F}(g^{-1}V)$. Reference: Stacks, tag 01E9.*

Lean: `TopCat.Presheaf.isFlasque_pushforward`
Uses: `def:flasque`
[absent from Mathlib: to be built]

Lemma 9.4 (Flasque sheaves are acyclic). *If $\mathcal{F}$ is flasque then $H^i(X, \mathcal{F}) = 0$ and $R^i f_* \mathcal{F} = 0$ for all $i > 0$. Reference: Stacks, Lemma 20.12.3, tag 01EB; Hartshorne, III.2.5.*

Lean: `AlgebraicGeometry.Scheme.higherDirectImage_flasque_eq_zero`
Uses: `def:flasque`, `lem:les`, `lem:hdi-sheafification`
[absent from Mathlib: to be built]

Proof sketch. Flasqueness is preserved under quotients by flasque subsheaves, and if two terms of a short exact sequence are flasque so is the third; embedding $\mathcal{F}$ in an injective (hence flasque) $\mathcal{I}$ and using the long exact sequence (Lemma 8.4) gives the vanishing by dimension shifting. For $R^i f_*$ use Lemma 8.5 and the same argument on each $f^{-1}(V)$. Faithful to Stacks, tag 01EB. $\square$

Lemma 9.5 (Acyclic resolutions compute derived functors). *If $0 \rightarrow \mathcal{F} \rightarrow \mathcal{A}^0 \rightarrow \mathcal{A}^1 \rightarrow \dots$ is a resolution with each $\mathcal{A}^k$ acyclic for $\Gamma(X, -)$ (resp. f_*), then $H^i(X, \mathcal{F}) \cong H^i(\Gamma(X, \mathcal{A}^\bullet))$ (resp. $R^i f_* \mathcal{F} \cong H^i(f_* \mathcal{A}^\bullet)$). Reference: Stacks, tag 0156 (derived functor via acyclic resolution); Hartshorne, III.1.2A.*

Lean: `AlgebraicGeometry.Scheme.cohomology_of_acyclicResolution`

Uses: `lem:flasque-acyclic`

[absent from Mathlib: to be built]

Lemma 9.6 (Acyclic pushforward lemma). *Let $X \xrightarrow{g} X' \xrightarrow{f} Y$ and let $\mathcal{B}$ be an $\mathcal{O}_X$ -module with $R^q g_* \mathcal{B} = 0$ for all $q > 0$. Then $R^q f_*(g_* \mathcal{B}) \cong R^q(f \circ g)_* \mathcal{B}$ naturally for all $q \geq 0$. Reference: the note [7].*

Lean: `AlgebraicGeometry.Scheme.higherDirectImage_pushforward_of_acyclic`

Uses: `lem:injective-flasque`, `lem:pushforward-flasque`, `lem:flasque-acyclic`, `lem:acyclic-resolution`

[absent from Mathlib: to be built]

Proof. Take an injective resolution $\mathcal{B} \rightarrow I^\bullet$. Each I^k is flasque (Lemma 9.2); since $R^q g_* \mathcal{B} = H^q(g_* I^\bullet)$, the hypothesis says $g_* I^\bullet$ is a resolution of $g_* \mathcal{B}$, with terms flasque (Lemma 9.3) hence f_* -acyclic (Lemma 9.4). By Lemma 9.5 and $f_* \circ g_* = (f \circ g)_*$, $R^q f_*(g_* \mathcal{B}) = H^q(f_* g_* I^\bullet) = H^q((f \circ g)_* I^\bullet) = R^q(f \circ g)_* \mathcal{B}$. Faithful to [7]. $\square$

Chapter 10

Comparison: abstract `rightDerived` vs sheaf cohomology

Proposition 10.1 (The abstract derived pushforward computes $R^n f_*$). *For $f: X \rightarrow Y$ and a quasi-coherent $M \in \text{SheafOfModules}(X)$ there is a natural isomorphism $((\text{Modules.pushforward } f).\text{rightDerived } n)(M) \cong R^n f_* M$, compatible with the equivalence between geometric $\mathcal{O}_X$ -modules and SheafOfModules . In particular the object in Theorem 1.1 is the geometric higher direct image.*

Lean: `AlgebraicGeometry.Scheme.Modules.rightDerivedPushforward_iso_higherDirectImage`

Uses: `def:higher-direct-image`, `lem:pf-left-exact`, `lem:flasque-acyclic`

[absent from Mathlib: to be built]

Proof strategy. Both are δ -functors agreeing in degree 0 (f_*); it suffices to show the abstract one is erasable/effaceable, i.e. computed by a class of objects acyclic on both sides. Injective objects of `SheafOfModules` map to flasque $\mathcal{O}_X$ -modules (Lemma 9.2 adapted), which are f_* -acyclic (Lemma 9.4); by uniqueness of derived functors the two δ -functors are isomorphic. This is the single comparison tying the categorical machinery of the target statement to the geometric development. $\square$

Part IV

Čech cohomology and vanishing on affines

References: Stacks, Sections 20.9–20.11 and 30.2–30.4 [8]; Hartshorne, III.3–III.4 [3].

Chapter 11

The Čech complex

Definition 11.1 (Čech complex and cohomology). For an open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of X and an $\mathcal{O}_X$ -module $\mathcal{F}$, the Čech complex is $\check{C}^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0 \dots i_p})$ with the usual alternating differential, and $\check{H}^p(\mathcal{U}, \mathcal{F})$ its cohomology. Reference: Stacks, tag [01ED](#); Hartshorne, III.4.

Lean: `AlgebraicGeometry.Scheme.cekComplex`, `AlgebraicGeometry.Scheme.cekCohomology`
[absent from Mathlib: to be built]

Lemma 11.2 (Čech degree 0 and functoriality). $\check{H}^0(\mathcal{U}, \mathcal{F}) \cong \Gamma(X, \mathcal{F})$, and a short exact sequence of sheaves induces a long exact sequence of Čech cohomology when the cover consists of sheafy-acyclic pieces (e.g. affines with quasi-coherent $\mathcal{F}$). Reference: Stacks, tags [01EF](#), [01EO](#).

Lean: `AlgebraicGeometry.Scheme.cekCohomology_zero`
Uses: `def:cek`
[absent from Mathlib: to be built]

Proposition 11.3 (Čech-to-derived comparison). There is a natural map $\check{H}^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$, an isomorphism in degrees 0, 1 in general, and an isomorphism in all degrees when every finite intersection $U_{i_0 \dots i_p}$ is $\mathcal{F}$ -acyclic (Leray's theorem / acyclic covers). Reference: Stacks, Lemma 20.11.6, tag [01EW](#); Hartshorne, III.4.5.

Lean: `AlgebraicGeometry.Scheme.cekToDerived`, `AlgebraicGeometry.Scheme.cekToDerived_iso_of_acyclicCover`
Uses: `def:cek`, `def:sheaf-cohomology`, `lem:flasque-acyclic`
[absent from Mathlib: to be built]

Chapter 12

Vanishing on affines and the comparison

Theorem 12.1 (Serre: vanishing on affines). *If $X = \operatorname{Spec} R$ is affine and $\mathcal{F}$ is quasi-coherent, then $H^i(X, \mathcal{F}) = 0$ for all $i > 0$. Reference: Stacks, Lemma 30.2.2, tag [01XB](#); Hartshorne, III.3.5 (Noetherian case) and III.3.7.*

Lean: `AlgebraicGeometry.cohomology_qcoh_affine_eq_zero`

Uses: `def:qcoh`, `prop:affine-equiv`, `def:sheaf-cohomology`, `prop:cech-to-derived`

[absent from Mathlib: to be built]

Proof outline. By Lemma [8.5/11.3](#) reduce to showing the Čech cohomology of $\widetilde{M}$ for the standard cover of $\operatorname{Spec} R$ by basic opens $D(f_i)$ (with $\sum(f_i) = R$) vanishes above degree 0; the Čech complex is the Koszul-type complex computing $\check{H}^p = 0$ for $p > 0$ because localisation is exact and the f_i generate the unit ideal. In the Noetherian case one may instead use that a quasi-coherent sheaf on $\operatorname{Spec} R$ embeds in a flasque quasi-coherent sheaf and induct. Faithful to Stacks, tag [01XB](#). $\square$

Corollary 12.2 (Affine covers compute cohomology). *If X is separated (so intersections of affine opens are affine) and $\mathcal{U}$ is an affine open cover, then for quasi-coherent $\mathcal{F}$, $\check{H}^p(\mathcal{U}, \mathcal{F}) \cong H^p(X, \mathcal{F})$ for all p . Reference: Stacks, Lemma 30.2.6, tag [01X9](#); Hartshorne, III.4.5.*

Lean: `AlgebraicGeometry.Scheme.cechCohomology_iso_cohomology_of_separated`

Uses: `prop:cech-to-derived`, `thm:affine-vanishing`

[absent from Mathlib: to be built]

Chapter 13

Higher direct images of quasi-coherent sheaves

Proposition 13.1 ($R^i f_*$ preserves quasi-coherence). *If $f: X \rightarrow Y$ is quasi-compact and quasi-separated and $\mathcal{F}$ is quasi-coherent, then $R^i f_* \mathcal{F}$ is quasi-coherent for all $i \geq 0$. Reference: Stacks, Lemma 30.4.5, tag 01XJ.*

Lean: AlgebraicGeometry.Scheme.higherDirectImage_isQuasiCoherent

Uses: def:higher-direct-image, lem:hdi-sheafification, thm:affine-vanishing, prop:qcoh-abelian

[absent from Mathlib: to be built]

Proof outline. By Lemma 8.5 the question is local on Y , so take $Y = \operatorname{Spec} A$. Cover X by finitely many affines; by Corollary 12.2 the Čech complex of quasi-coherent sheaves computes $R^i f_* \mathcal{F}$, and each Čech term is quasi-coherent (pushforward of quasi-coherent along the affine inclusions of intersections, using *qcqs*); quasi-coherence is closed under kernels/cokernels (Proposition 6.5), so the cohomology is quasi-coherent. Faithful to Stacks, tag 01XJ. $\square$

Proposition 13.2 (Direct image over an affine base). *For A a ring, $f: X \rightarrow \operatorname{Spec} A$ quasi-compact quasi-separated and $\mathcal{F}$ quasi-coherent, $R^i f_* \mathcal{F} \cong \widetilde{H^i(X, \mathcal{F})}$; in particular $R^i f_* \mathcal{F}$ is coherent iff $H^i(X, \mathcal{F})$ is a finite A -module. Reference: Stacks, Lemma 30.4.6, tag 01XX.*

Lean: AlgebraicGeometry.Scheme.higherDirectImage_over_affine

Uses: prop:hdi-qcoh, prop:affine-equiv, lem:tilde-sections

[absent from Mathlib: to be built]

Part V

Projective space and Proj

References: Stacks, Sections 27.10–27.14 (Proj) and 30.8, 30.14–30.15 [8]; Hartshorne, II.5, III.5 [3]; Serre [5].

Chapter 14

Twisting sheaves and graded modules on Proj

Construction 14.1 (The functor $\widetilde{(-)}$ on Proj and twists). Let S be a graded ring, finitely generated in degree 1 over $S_0 = A$, and $X = \text{Proj } S$ (a scheme in Mathlib). For a graded S -module M there is a quasi-coherent sheaf $\widetilde{M}$ on X ; taking $M = S(d)$ defines the twisting sheaves $\mathcal{O}_X(d) = \widetilde{S(d)}$, with $\mathcal{O}_X(d) \otimes \mathcal{O}_X(e) \cong \mathcal{O}_X(d+e)$ and $\mathcal{O}_X(d)$ invertible when S is generated in degree 1. Write $\mathcal{F}(d) = \mathcal{F} \otimes \mathcal{O}_X(d)$. Reference: Stacks, Section 27.10, tags [01M3](#), [01MT](#); Hartshorne, II.5.

Lean: `AlgebraicGeometry.Proj.tildeGraded`, `AlgebraicGeometry.Proj.twistingSheaf`

Uses: `def:qcoh`, `def:invertible`

[absent from Mathlib: to be built]

Lemma 14.2 (Sections on the standard cover). For $f \in S_1$, $D_+(f) = \text{Spec}(S_{(f)})$ and $\mathcal{O}_X(d)|_{D_+(f)} \cong \widetilde{(S(d)_{(f)})}$; the standard cover $\{D_+(x_i)\}$ for degree-1 generators x_i is affine, and its intersections are the $D_+(x_{i_0} \cdots x_{i_p})$. Reference: Stacks, tag [01M3](#).

Lean: `AlgebraicGeometry.Proj.twistingSheaf_restrict`

Uses: `con:proj-tilde`, `lem:tilde-sections`

[absent from Mathlib: to be built]

Proposition 14.3 (Every coherent sheaf on Proj comes from a graded module). Let A be Noetherian and S a finitely generated graded A -algebra generated in degree 1, $X = \text{Proj } S$. Every coherent $\mathcal{O}_X$ -module is isomorphic to $\widetilde{M}$ for a finitely generated graded S -module M , and there is d_0 with $M_d \xrightarrow{\sim} H^0(X, \mathcal{F}(d))$ for $d \geq d_0$. Reference: Stacks, Proposition 30.14.4 and Section 30.15, tags [01YS](#), [01YW](#); Hartshorne, II.5.15–5.17.

Lean: `AlgebraicGeometry.Proj.coherent_from_gradedModule`

Uses: `con:proj-tilde`, `def:coherent`, `lem:proj-sections`

[absent from Mathlib: to be built]

Chapter 15

Cohomology of projective space

Theorem 15.1 (Serre’s computation). *Let A be a ring, $n \geq 0$, $\mathbf{P}_A^n = \text{Proj } A[T_0, \dots, T_n]$. Then, as graded A -modules,*

$$\bigoplus_d H^0(\mathbf{P}_A^n, \mathcal{O}(d)) \cong A[T_0, \dots, T_n], \quad \bigoplus_d H^n(\mathbf{P}_A^n, \mathcal{O}(d)) \cong \frac{1}{T_0 \cdots T_n} A[\frac{1}{T_0}, \dots, \frac{1}{T_n}],$$

and $H^q(\mathbf{P}_A^n, \mathcal{O}(d)) = 0$ for $0 < q < n$; each $H^q(\mathbf{P}_A^n, \mathcal{O}(d))$ is a finite free A -module. Reference: Stacks, Lemma 30.8.1, tag [01XS](#); Hartshorne, III.5.1.

Lean: AlgebraicGeometry.cohomology_projectiveSpace_twist

Uses: thm:affine-vanishing, cor:affine-cover-computes, lem:proj-sections

[absent from Mathlib: to be built]

Proof outline. The standard cover $\{D_+(T_i)\}$ is affine with affine intersections, so by Corollary 12.2 the cohomology is that of the Čech complex $\check{C}^\bullet(\{D_+(T_i)\}, \mathcal{O}(d))$. This complex is, in each graded piece, an explicit Koszul-type complex of localisations of the polynomial ring; a direct computation of its cohomology gives H^0 (degree d polynomials), H^n (the stated “dual” piece), and vanishing in between. Finiteness/freeness over A is read off the monomial bases. Faithful to Stacks, tag [01XS](#). $\square$

Corollary 15.2 (Finiteness and Serre vanishing on $\mathbf{P}^n$). *$H^q(\mathbf{P}_A^n, \mathcal{O}(d))$ is a finite A -module for all q, d , and vanishes for $q > 0$ and $d \gg 0$.*

Lean: AlgebraicGeometry.cohomology_projectiveSpace_finite_and_vanishing

Uses: thm:cohomology-Pn

[absent from Mathlib: to be built]

Chapter 16

Serre finiteness and vanishing on Proj

Lemma 16.1 (Affine finiteness, projective case). *Let A be Noetherian and $X = \text{Proj } S$ projective over $\text{Spec } A$ (with S as in Proposition 14.3), $\mathcal{F}$ coherent. Then $H^i(X, \mathcal{F})$ is a finite A -module for all i , and $H^i(X, \mathcal{F}(d)) = 0$ for $i > 0$, $d \gg 0$; moreover $\mathcal{F}(d)$ is globally generated for $d \gg 0$. Reference: Stacks, Lemma 30.15.2 and Section 30.15, tag 01YW; Hartshorne, III.5.2.*

Lean: `AlgebraicGeometry.Proj.cohomology_coherent_finite_and_vanishing`

Uses: `thm:cohomology-Pn`, `prop:proj-graded`, `def:coherent`

[absent from Mathlib: to be built]

Proof outline. Write $\mathcal{F} = \widetilde{M}$ (Proposition 14.3) and present M as a quotient of a finite free graded module; this yields a finite resolution of $\mathcal{F}$ by finite direct sums of twists $\mathcal{O}_X(a_j)$ up to a coherent kernel, and $H^\bullet$ of each such sum is finite over A by Theorem 15.1. Descending induction on the (finite) cohomological amplitude, using the long exact sequence (Lemma 8.4), gives finiteness of $H^i(X, \mathcal{F})$ and, after twisting by $\mathcal{O}(d)$ with $d \gg 0$, the vanishing and global generation. Faithful to Stacks, tag 01YW. $\square$

Part VI

Coherent sheaves

References: Stacks, Sections 30.9, 30.11, 30.12 [\[8\]](#); Hartshorne, II.5 [\[3\]](#).

Chapter 17

Coherence on locally Noetherian schemes

Definition 17.1 (Coherent module). An $\mathcal{O}_X$ -module $\mathcal{F}$ is *coherent* if it is of finite type and, for every open U and every morphism $\mathcal{O}_U^m \rightarrow \mathcal{F}|_U$, the kernel is of finite type. Reference: Stacks, tag [01BU](#).

Lean: `AlgebraicGeometry.Scheme.Modules.IsCoherent`

Uses:

[absent from Mathlib: to be built]

Lemma 17.2 (Locally Noetherian source). *If $f: X \rightarrow S$ is of finite type and S is locally Noetherian, then X is locally Noetherian. In particular a proper f over locally Noetherian S has locally Noetherian source. Reference: Stacks, Lemma 29.15.6, tag [01T6](#).*

Lean: `AlgebraicGeometry.locallyNoetherian_of_finiteType`

[partly in Mathlib: must be extended]

Lemma 17.3 (Coherence on locally Noetherian schemes). *On a locally Noetherian scheme X , an $\mathcal{O}_X$ -module is coherent iff it is quasi-coherent and of finite type. Reference: Stacks, Lemma 30.9.1, tag [01XZ](#).*

Lean: `AlgebraicGeometry.Scheme.Modules.isCoherent_iff_qcoh_finiteType`

Uses: `def:coherent`, `def:qcoh`

[absent from Mathlib: to be built]

Lemma 17.4 (Finite presentation = coherence). *On a locally Noetherian scheme, an $\mathcal{O}_X$ -module is of finite presentation (in the sense of `SheafOfModules.IsFinitePresentation`) iff it is coherent. Reference: Stacks, tag [01XZ](#) together with tag [01BN](#).*

Lean: `AlgebraicGeometry.Scheme.Modules.isFinitePresentation_iff_isCoherent`

Uses: `lem:coherent-noeth`, `lem:qcoh-agree`

[absent from Mathlib: to be built]

Proposition 17.5 (Coherent sheaves form a Serre–abelian subcategory). *On a locally Noetherian scheme, kernels, cokernels, images and extensions of coherent sheaves are coherent; in a short exact sequence, if two terms are coherent so is the third. Reference: Stacks, Lemmas 30.9.2, 30.9.3, tags [01Y0](#), [01Y1](#).*

Lean: `AlgebraicGeometry.Scheme.Modules.isCoherent_of_ses`

Uses: `def:coherent`, `lem:coherent-noeth`, `prop:qcoh-abelian`

[absent from Mathlib: to be built]

Chapter 18

Closed immersions and support

Lemma 18.1 (Pushforward along a closed immersion). *Let $i: Z \rightarrow X$ be a closed immersion of locally Noetherian schemes and $\mathcal{G}$ a coherent $\mathcal{O}_Z$ -module. Then $i_*\mathcal{G}$ is coherent, i_* is exact, and $R^pi_*\mathcal{G} = 0$ for all $p > 0$. Reference: Stacks, Lemma 30.9.9, tag [01Y6](#); Liu, 5.1.14(d) [6].*

Lean: `AlgebraicGeometry.Scheme.closedImmersion_pushforward_coherent_and_acyclic`

Uses: `def:coherent`, `def:higher-direct-image`, `lem:flasque-acyclic`

[absent from Mathlib: to be built]

Proof sketch. Coherence of $i_*\mathcal{G}$ is local and reduces to the affine statement that a finite module over R/I is finite over R . Exactness and the vanishing of R^pi_* for $p > 0$ hold because i is a homeomorphism onto a closed subset, so i_* is exact on the underlying sheaves and i_* of a flasque sheaf is flasque, hence acyclic (Lemma 9.4). Faithful to Stacks, tag [01Y6](#). $\square$

Lemma 18.2 (Support and generic stalks). *For a coherent $\mathcal{F}$ on a Noetherian scheme, $\text{Supp}(\mathcal{F})$ is closed; if $Z \subseteq X$ is integral closed with generic point ξ then $\mathcal{O}_{X,\xi} = \kappa(\xi)$ is a field, so any coherent $\mathcal{G}$ supported on Z has $\mathcal{G}_\xi$ a finite-dimensional $\kappa(\xi)$ -vector space, and $\mathcal{G}_\xi$ is automatically annihilated by $\mathfrak{m}_\xi = 0$. Reference: Stacks, tags [01B4](#), [01YB](#).*

Lean: `AlgebraicGeometry.Scheme.Modules.support_closed`, `AlgebraicGeometry.Scheme.Modules.genericStalk_finrank`

Uses: `def:coherent`

[absent from Mathlib: to be built]

Part VII

Projective morphisms

References: Stacks, Sections 29.40 (ample), 30.16 [\[8\]](#); Hartshorne, II.7, III.8.8 [\[3\]](#).

Chapter 19

Ample invertible sheaves and projective morphisms

Definition 19.1 (Relatively ample; projective morphism). An invertible $\mathcal{O}_X$ -module $\mathcal{L}$ is *f-relatively ample* (for $f: X \rightarrow S$) if, locally on $S = \bigcup \operatorname{Spec} A_i$, $X_i \rightarrow \operatorname{Spec} A_i$ is isomorphic over A_i to a locally closed immersion into some $\mathbf{P}_{A_i}^{n_i}$ pulling $\mathcal{O}(1)$ back to $\mathcal{L}$. A morphism $f: X \rightarrow S$ is *projective* if it factors as a closed immersion $X \hookrightarrow \mathbf{P}_S^n$ followed by the projection (equivalently, admits an *f*-ample $\mathcal{L}$ with *f* proper). Reference: Stacks, Section 29.40, tag [01VG](#), and tag [01W8](#); Hartshorne, II.7.

Lean: `AlgebraicGeometry.Scheme.IsRelativelyAmple`, `AlgebraicGeometry.IsProjective`

Uses: `def:invertible`

[absent from Mathlib: to be built]

Lemma 19.2 (Pullback of $\mathcal{O}(1)$ is relatively ample). *If $i: X \rightarrow \mathbf{P}_S^n$ is a closed immersion over S then $i^*\mathcal{O}_{\mathbf{P}_S^n}(1)$ is *f-relatively ample*, where $f: X \rightarrow S$; moreover f is projective and proper. Reference: Stacks, Lemma 29.40.7, tag [02NR](#).*

Lean: `AlgebraicGeometry.Scheme.relativelyAmple_pullback_twist`

Uses: `def:projective-morphism`

[absent from Mathlib: to be built]

Chapter 20

Higher direct images along projective morphisms

Lemma 20.1 (Graded finiteness along a projective morphism). *Let S be Noetherian affine, $f: X \rightarrow \operatorname{Spec} A$ projective with f -relatively ample $\mathcal{L}$, and $\mathcal{F}$ coherent. Then $\bigoplus_d H^p(X, \mathcal{F} \otimes \mathcal{L}^{\otimes d})$ is a finite graded module over the section ring, for each p ; in particular each $H^p(X, \mathcal{F})$ is a finite A -module and $H^p(X, \mathcal{F} \otimes \mathcal{L}^{\otimes d}) = 0$ for $p > 0$, $d \gg 0$. Reference: *Stacks*, Lemma 30.16.1 and Lemma 30.19.3, tags [0200](#), [0897](#).*

Lean: `AlgebraicGeometry.Scheme.gradedCohomology_finite_projective`

Uses: `lem:affine-finite-proj`, `prop:proj-graded`, `def:projective-morphism`

[absent from Mathlib: to be built]

Lemma 20.2 (Vanishing after a large relatively ample twist). *Let S be Noetherian, $f: X \rightarrow S$ proper, $\mathcal{L}$ an f -relatively ample invertible sheaf, $\mathcal{F}$ coherent. Then there is n_0 with $R^p f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes n}) = 0$ for all $p > 0$ and $n \geq n_0$. Reference: *Stacks*, Lemma 30.16.2, tag [0201](#).*

Lean: `AlgebraicGeometry.Scheme.higherDirectImage_vanishing_relativelyAmple`

Uses: `lem:graded-finiteness`, `prop:hdi-affine-base`, `lem:hdi-sheafification`

[absent from Mathlib: to be built]

Proof sketch. The statement is local on S (Lemma [8.5](#)), so take $S = \operatorname{Spec} A$ Noetherian. By Proposition [13.2](#), $R^p f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes n}) = H^p(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n})$, which vanishes for $p > 0$, $n \gg 0$ by the Serre vanishing in Lemma [20.1](#). Faithful to *Stacks*, tag [0201](#). $\square$

Lemma 20.3 (Coherence in the projective case). *Let S be locally Noetherian, $f: X \rightarrow S$ projective, $\mathcal{F}$ coherent. Then $R^p f_* \mathcal{F}$ is coherent for all $p \geq 0$. Reference: *Stacks*, Lemma 30.16.3, tag [0204](#); *Hartshorne*, III.8.8.*

Lean: `AlgebraicGeometry.Scheme.higherDirectImage_coherent_of_projective`

Uses: `lem:graded-finiteness`, `prop:hdi-affine-base`, `lem:coherent-noeth`

[absent from Mathlib: to be built]

Proof sketch. Local on S , reduce to $S = \operatorname{Spec} A$ Noetherian; then $R^p f_* \mathcal{F} = \widetilde{H^p(X, \mathcal{F})}$ (Proposition [13.2](#)), and $H^p(X, \mathcal{F})$ is a finite A -module by Lemma [20.1/16.1](#), so $R^p f_* \mathcal{F}$ is coherent (Lemma [17.3](#)). Faithful to *Stacks*, tag [0204](#). $\square$

Part VIII

Dévissage

References: Stacks, Section 30.12 [\[8\]](#).

Chapter 21

Dévissage of coherent sheaves

Lemma 21.1 (Generation by pushforwards from integral subschemes). *Let X be Noetherian and $\mathcal{P}$ a property of coherent sheaves that holds for $(Z \hookrightarrow X)_*\mathcal{I}$ for every integral closed $Z \subseteq X$ and every coherent ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_Z$. Then $\mathcal{P}$ holds for all coherent sheaves. Reference: Stacks, Lemmas 30.12.4, 30.12.5, tags [01YG](#), [01YH](#).*

Lean: `AlgebraicGeometry.Scheme.devissage_generation`

Uses: `def:coherent`, `prop:coherent-abelian`

[absent from Mathlib: to be built]

Theorem 21.2 (Dévissage). *Let X be Noetherian and $\mathcal{P}$ a property of coherent $\mathcal{O}_X$ -modules such that (1) in any short exact sequence, if two terms satisfy $\mathcal{P}$ so does the third; and (2) for every integral closed $Z \subseteq X$ with generic point ξ there is a coherent $\mathcal{G}$ with $\text{Supp } \mathcal{G} = Z$, $\dim_{\kappa(\xi)} \mathcal{G}_\xi = 1$, and $\mathcal{P}(\mathcal{G})$. Then $\mathcal{P}$ holds for every coherent $\mathcal{O}_X$ -module. Reference: Stacks, Lemma 30.12.6, tag [01YI](#).*

Lean: `AlgebraicGeometry.Scheme.coherent_devissage`

Uses: `lem:devissage-gen`, `prop:coherent-abelian`, `lem:support`

[absent from Mathlib: to be built]

Proof sketch. By Lemma [21.1](#) it suffices to treat $(Z \hookrightarrow X)_*\mathcal{I}$. If $\mathcal{P}$ failed for some such sheaf, Noetherian induction on the support gives a minimal integral closed Z_0 on which it fails, while it holds on every proper closed subscheme of Z_0 ; hypothesis (2) for Z_0 and the 2-out-of-3 property (1) applied to a suitable short exact sequence relating $\mathcal{I}$, $\mathcal{G}$ and sheaves supported on lower-dimensional closed subsets yield a contradiction. Faithful to Stacks, tag [01YI](#). $\square$

Part IX

Chow's lemma

References: Stacks, Section 30.18 [\[8\]](#); EGA II, 5.6.1 [\[1\]](#).

Chapter 22

Chow's lemma

Theorem 22.1 (Chow's lemma). *Let S be Noetherian and $f: X \rightarrow S$ separated of finite type. Then there exist n , a proper surjective $\pi: X' \rightarrow X$ that is an isomorphism over a dense open $U \subseteq X$, and an immersion $X' \hookrightarrow \mathbf{P}_S^n$ over S . If f is proper then $X' \rightarrow S$ is projective. Reference: Stacks, Lemma 30.18.1, tag [0200](#).*

Lean: AlgebraicGeometry.chowLemma

Uses: def:projective-morphism

[absent from Mathlib: to be built]

Proof outline. Cover X by finitely many affine opens U_i containing all generic points, so $U = \bigcap U_i$ is dense; embed each U_i as a scheme-theoretically dense open of a closed subscheme $Z_i \subseteq \mathbf{P}_S^{n_i}$. Let Z be the scheme-theoretic image of $U \rightarrow \prod_S Z_i$; the projections give proper $p_i: Z \rightarrow Z_i$, and gluing the $p_i^{-1}(U_i)$ produces $\pi: X' \rightarrow X$ proper and surjective, an isomorphism over U , with X' placed in one $\mathbf{P}_S^n$ by the Segre embedding. Faithful to Stacks, tag [0200](#). $\square$

Remark 22.2 (The graph is a closed immersion; ampleness). If $i: X' \rightarrow \mathbf{P}_S^m$ is the immersion and $\pi: X' \rightarrow X$ proper, then $(i, \pi): X' \rightarrow \mathbf{P}_X^m$ is a closed immersion, so $\mathcal{L} := i^* \mathcal{O}(1) \cong (i, \pi)^* \mathcal{O}_{\mathbf{P}_X^m}(1)$ is π -relatively ample; and $X' \rightarrow S$ is projective. Reference: Stacks, Remark 30.18.2, tag [0201](#); Lemma [19.2](#).

Lean: AlgebraicGeometry.chowLemma_graph_closedImmersion

Uses: thm:chow, lem:pullback-ample

[absent from Mathlib: to be built]

Part X

The finiteness theorem

Chapter 23

Assembly

Theorem 23.1 (Coherence of higher direct images). *Let S be locally Noetherian, $f: X \rightarrow S$ proper, $\mathcal{F}$ coherent. Then $R^p f_* \mathcal{F}$ is coherent for all $p \geq 0$. Reference: Stacks, Proposition 30.19.1, tag 0205; EGA III, 3.2.1.*

Lean: AlgebraicGeometry.Scheme.higherDirectImage_coherent_of_isProper

Uses: thm:devissage, thm:chow, rem:chow-graph, lem:proj-hdi-coherent, lem:proj-hdi-vanish, lem:acyclic-pushforward, lem:closed-immersion-acyclic, prop:coherent-abelian, prop:hdi-qcoh, lem:X-noeth, lem:support

[absent from Mathlib: to be built]

Proof. Coherence of $R^p f_* \mathcal{F}$ is local on S (Lemma 8.5 and Proposition 13.1), so assume S Noetherian; then X is Noetherian (Lemma 17.2). Put $\mathcal{P}(\mathcal{F}) \Leftrightarrow “R^p f_* \mathcal{F} \text{ coherent for all } p”$.

Hypothesis (1) of Theorem 21.2. For a short exact sequence of coherent sheaves, the long exact sequence of the $R^p f_*$ (Lemma 8.4) places each $R^p f_*$ of the third sheaf between two coherent terms; by Proposition 17.5 (2-out-of-3) it is coherent. So $\mathcal{P}$ satisfies 2-out-of-3.

Hypothesis (2). Fix an integral closed $Z \subseteq X$ with generic point ξ , closed immersion $j: Z \rightarrow X$, and $g = f \circ j: Z \rightarrow S$. It suffices to produce coherent $\mathcal{G}$ on Z with (a) $\dim_{\kappa(\xi)} \mathcal{G}_\xi = 1$ and (b) $R^p g_* \mathcal{G}$ coherent for all p : then $\mathcal{F}_0 := j_* \mathcal{G}$ is coherent (Lemma 18.1), and since $R^q j_* \mathcal{G} = 0$ for $q > 0$ the acyclic pushforward Lemma 9.6 gives $R^p f_* \mathcal{F}_0 \cong R^p (f \circ j)_* \mathcal{G} = R^p g_* \mathcal{G}$, coherent by (b), so $\mathcal{P}(\mathcal{F}_0)$; and $\text{Supp } \mathcal{F}_0 = Z$, $(\mathcal{F}_0)_\xi = \mathcal{G}_\xi$ is 1-dimensional, matching Theorem 21.2(2).

Construct $\mathcal{G}$ by Chow’s lemma (Theorem 22.1) applied to $g: Z \rightarrow S$: a proper surjective $\pi: Z' \rightarrow Z$, isomorphism over a dense open $U \subseteq Z$, with $g' = g \circ \pi: Z' \rightarrow S$ projective and an invertible $\mathcal{L}$ on Z' that is π -relatively ample (Remark 22.2). By Lemma 20.2 choose $n \geq 1$ with $R^q \pi_* \mathcal{L}^{\otimes n} = 0$ for $q > 0$, and set $\mathcal{G} = \pi_* \mathcal{L}^{\otimes n}$. Then:

- $\mathcal{G}$ is coherent, being $\pi_* \mathcal{L}^{\otimes n} = R^0 \pi_* \mathcal{L}^{\otimes n}$ for the projective π (Lemma 20.3);
- (a) holds: over U , π is an isomorphism so $\mathcal{G}|_U = \mathcal{L}^{\otimes n}|_U$ is invertible, giving $\text{Supp } \mathcal{G} = Z$ and $\dim_{\kappa(\xi)} \mathcal{G}_\xi = 1$ (Lemma 18.2);
- (b) holds by Lemma 9.6 applied to $Z' \xrightarrow{\pi} Z \xrightarrow{g} S$ with $\mathcal{B} = \mathcal{L}^{\otimes n}$: as $R^q \pi_* \mathcal{L}^{\otimes n} = 0$ for $q > 0$, $R^p g_* \mathcal{G} \cong R^p g'_* \mathcal{L}^{\otimes n}$, coherent for all p because g' is projective (Lemma 20.3).

By Theorem 21.2, $\mathcal{P}(\mathcal{F})$ holds for all coherent $\mathcal{F}$. Only the single vanishing $R^q \pi_* \mathcal{L}^{\otimes n} = 0$ ($q > 0$) is used, coherence being transported through Lemma 9.6; no spectral sequence appears (Remark 3.1). Faithful to Stacks, tag 0205. $\square$

Part XI

The target Lean statement

Chapter 24

From coherence to finite presentation

Theorem 24.1 (Target). *Let $f: X \rightarrow Y$ be proper, Y locally Noetherian, M an $\mathcal{O}_X$ -module of finite presentation. Then for all n , $((\text{Modules.pushforward } f).\text{rightDerived } n)(M)$ is of finite presentation. This is Theorem 1.1.*

Lean: AlgebraicGeometry.Scheme.rightDerivedPushforward_isFinitePresentation_of_isProper

Uses: thm:finiteness, prop:bridge-rderived, lem:fp-iff-coh, lem:X-noeth

[absent from Mathlib: to be built]

Proof. f proper $\Rightarrow$ of finite type, so X is locally Noetherian (Lemma 17.2); by Lemma 17.4, M is coherent. By Proposition 10.1 the derived object is $R^n f_* M$, coherent by Theorem 23.1; applying Lemma 17.4 on the locally Noetherian base Y returns finite presentation. $\square$

Part XII

Dependency graph, critical path, and work packages

Chapter 25

Critical path and packaging

25.1 Critical path to Theorem 23.1

Ch.5 tilde $\Rightarrow$ Ch.6 QCoh $\Rightarrow$ Ch.8 cohomology
 $\Rightarrow$ Ch.11–12 Čech + affine vanishing $\Rightarrow$ Ch.13 qcoh of $R^i f_*$
 $\Rightarrow$ Ch.15 coh of $\mathbf{P}^n \Rightarrow$ Ch.20 projective case
and independently Ch.9 flasque, Ch.21 dévissage, Ch.22 Chow
 $\Rightarrow$ Ch.23 finiteness $\Rightarrow$ Ch.24 target.

25.2 Suggested work packages (parallelisable)

WP1 Affine QCoh. Ch.5–6: complete the $\widetilde{(-)}$ API in `Modules.Tilde`, the affine equivalence, and the geometric quasi-coherence predicate. Prerequisite for everything.

WP2 Cohomology core. Ch.8–9: H^i , $R^i f_*$, δ -functor structure, sheafification description, and the flasque package. Independent of WP1 except Lemma 8.5.

WP3 Čech + vanishing. Ch.11–13: needs WP1 and WP2; yields Serre vanishing, affine-cover computation, and quasi-coherence of $R^i f_*$.

WP4 Projective space. Ch.14–16: needs WP3; the twisting sheaves and Theorem 15.1.

WP5 Coherence. Ch.17–18: needs WP1; the coherence predicate, finite-presentation equivalence, closed immersions, support.

WP6 Projective morphisms. Ch.19–20: needs WP4, WP5.

WP7 Dévissage. Ch.21: needs WP5. Parallel to WP6.

WP8 Chow. Ch.22: needs scheme-theoretic image and Segre; largely independent morphism-theory package.

WP9 Assembly + bridge. Ch.23–24: needs WP2 (flasque/acyclic pushforward and Prop.10.1), WP6, WP7, WP8.

25.3 Largest single items

By anticipated effort: Theorem 12.1 (Serre vanishing on affines) and the Čech comparison (WP3); Theorem 15.1 (explicit cohomology of $\mathbf{P}^n$, WP4); Theorem 22.1 (Chow’s lemma, WP8); and Proposition 10.1 (the `rightDerived`/sheaf-cohomology comparison, WP2). None requires a spectral sequence.

25.4 Items closest to existing Mathlib

Lemma 17.2, Lemma 8.3, Lemma 8.4 (via the `RightDerived` δ -functor API), and the tilde sections/stalks (Lemma 5.2) are the shortest to land, being extensions of declarations already present (Chapter 2).

Bibliography

- [1] A. Grothendieck, J. Dieudonné, EGA II, Publ. Math. IHÉS 8 (1961), Théorème 5.6.1 (Chow's lemma).
- [2] A. Grothendieck, J. Dieudonné, EGA III, Publ. Math. IHÉS 11 (1961), Théorème 3.2.1.
- [3] R. Hartshorne, Algebraic Geometry, GTM 52, Springer, 1977, Chapters II.5–7, III.1–5, III.8.
- [4] R. Vakil, The Rising Sea: Foundations of Algebraic Geometry, §18.9.
- [5] J.-P. Serre, Faisceaux algébriques cohérents, Ann. of Math. 61 (1955), 197–278.
- [6] Q. Liu, Algebraic Geometry and Arithmetic Curves, Oxford, 2002, 5.1.14.
- [7] Note on avoiding spectral sequences in EGA III.3.2.1 (attached).
- [8] The Stacks Project Authors, The Stacks Project, <https://stacks.math.columbia.edu>: Ch. 26 (tag [01HR](#)), Ch. 27 (tag [01M3](#)), Ch. 20 (tag [01DW](#)), Ch. 30 (tag [01X6](#)); Prop. 30.19.1 (tag [0205](#)).